

CE 509: Bridge Engineering

Course Code	CE 509
Course Name	Bridge Engineering
L-T-P-C	3-0-0-3
Prerequisites	CE 301 and CE 351
Intended for	B.Tech. 3rd/4th year/PG
Distribution	Elective
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)

Course Contents

- **Introduction**
Importance of bridges; Historical development and classification of bridge types; Various structural forms; Materials in bridge construction; Consideration for site selection; Determination of design discharge; linear waterways and economical span; High flood level (HFL) and vertical clearance; Traffic projection. [4 Lectures]
- **Design Loads**
Consideration of various forces and their application as per IRC specifications for loadings on road bridges. [3 Lectures]
- **Structural Behavior and Concepts for Analysis**
Moving load analysis; Grillage analogy; Structural behavior of box-girder bridges, arch bridges, suspension bridges, and cable stayed bridges. [8 Lectures]
- **Design of Bridge Superstructure**
Analysis and design of reinforced concrete slab bridge deck; Design considerations for skewed slab decks; Pre-tensioning and post-tensioning; Analysis and design of pre-stressed concrete slab deck; Introduction to T-beam bridges; Approximate methods for analysis of T-beam bridges: Courbon's method; Illustrative examples. [17 Lectures]
- **Bridge Bearings and Joints**
Importance and types of bearings; Design of elastomeric bearings; Seismic considerations for bearing design; Provisions for expansion joints. [5 Lectures]
- **Bridge Substructure**
Components of substructure; Computation of different forces acting on substructure; Introduction to various bridge foundation types. [3 Lectures]
- **Construction and Maintenance**
Common bridge construction techniques; overview of segmental construction for long span bridges; Primary [2 Lectures]

Text Books

1. D.J. Victor, **Essentials of Bridge Engineering**, 6th Edition, Oxford and F.H. Publishing, New Delhi, India, 2007
2. N.K. Raju, **Design of Bridges**, 4th Edition, Oxford and TBH Publishing, New Delhi, 2009.

References

1. E.C. Hambly, **Bridge Deck Behaviour**, 2nd Edition, CRC Press, Taylor and Francis Group, New York, USA, 1991.
2. IRC-112 (2011), Code of Practice for Concrete Road Bridges, Indian Roads Congress, New Delhi, India.
3. IRC-6 (2014), Standard Specifications and Code of Practice for Road Bridges, Section: II - Loads and Stresses, Indian Roads Congress, New Delhi, India.
4. N. Rajagopalan, **Bridge Superstructure**, Narosa Publishing House, New Delhi, India, 2013.
5. S. Ponnuswamy, **Bridge Engineering**, 2nd Edition, Tata McGraw-Hill Education, New Delhi, India, 2007.
6. V.K. Raina, **Raina's Concrete Bridge Practice Analysis, Design & Economics**, 4th Edition, Shroff Publishers and Distributors Pvt. Ltd., Mumbai, India, 2014.